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#include <iostream>

using namespace std;

int main() {
    int n = 4;
    int W = 7;

    int weight[] = {1, 3, 4, 5};
    int profit[] = {1, 4, 5, 7};

    int dp[5][8] = {0};

    for (int i = 1; i <= n; i++) {
        for (int w = 1; w <= W; w++) {

            if (weight[i - 1] <= w) {
                dp[i][w] = max(
                    profit[i - 1] + dp[i - 1][w - weight[i - 1]],
                    dp[i - 1][w]
                );
            }
            else {
                dp[i][w] = dp[i - 1][w];
            }
        }
    }

    int selected[4] = {0};
    int w = W;

    for (int i = n; i >= 1; i--) {

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        if (dp[i][w] != dp[i - 1][w]) {
            selected[i - 1] = 1;
            w = w - weight[i - 1];
        }
    }

    cout << "Maximum Profit = " << dp[n][W] << endl;

    cout << "Selected Items = ";

    for (int i = 0; i < n; i++) {
        cout << selected[i];
    }

    cout << endl;

    return 0;
}

```

OUTPUT:

Maximum Profit = 9

Selected Items = 0110

=== Code Execution Successful ===